Machine Learning Assignment 2

Topics: kNN, Logistic Regression and PCA

Total Marks: 10

Q1. Implement the k-Nearest Neighbours (kNN) classifier and evaluate its performance on the `Social Network Ads.csv` dataset, which is a categorical dataset used to determine whether a user purchased a product or not based on three features.

- Visualize the data using a three-dimensional plot of the features, using different colors to represent class labels 0 and 1.
- Implement the kNN classifier from scratch and compare its performance with the available library implementation. Use 90% of the data for training and the remaining 10% for testing the classification accuracy.
- Using the confusion matrix, compute the accuracy, precision, recall, and F1-score for different values of k.
- Apply the K-fold cross-validation technique (assume $K = 5$) and plot the average classification accuracy as a function of the number of neighbours k. Identify the value of k that yields the best accuracy.
- For the optimal value of k obtained above, compare the classification performance using the following distance or similarity metrics:
  - Euclidean distance
  - Cosine similarity
  - Mahalanobis distance

  Identify the metric that gives the best performance.
- From the above experiments, study the effect of extreme values of k (such as $k = 1$ and very large values of k). Infer the reasons behind the observed behavior. Repeat the experiment by fixing k and reducing the amount of training data (e.g., 40% and 60% of the dataset) to examine whether the same value of k performs well for different training set sizes.
- **Effect of k:** Visualize the decision boundaries by considering two features at a time for different values of k. Comment on which values of k result in smoother decision boundaries.
- **Regression:** Implement kNN for regression by considering the first two features as input variables and the third feature as the output variable. Use a 60:40 train–test split and report the mean squared error (MSE) on the test data for different values of k.

Q2. Perform binary classification using logistic regression on the dataset `Social Network Ads.csv`, which is a categorical dataset used to determine whether a user purchased a product or not based on three features.

1. Visualize the data by plotting the features in a three-dimensional space, using different colors to represent class labels 0 and 1.
2. Implement the logistic (sigmoid) function and learn the model parameters using the following optimization methods:
   - Stochastic gradient descent.
   - Newton's method (iteratively reweighted least squares method). Examine whether the IRLS method converges in fewer iterations compared to gradient descent.
   - An available library implementation. Compare the results with the above methods.

3. Plot the training data, test data, and the decision boundary learned by logistic regression in the same figure. The decision boundary should be a straight line separating the region where $h_{\mathbf{w}}(x) > 0.5$ from the region where $h_{\mathbf{w}}(x) \leq 0.5$, where

$$h_{\mathbf{w}}(x^{(i)}) = \sigma(\mathbf{w}^\top x^{(i)}).$$

Use 90% of the data for training and the remaining 10% for testing the classification performance. Using the confusion matrix, compute the accuracy, precision, recall, and F1-score for each method.

Q3. An imbalanced dataset refers to a dataset in which the classes are not represented equally. Classification problems involving multiple classes with imbalanced data present additional challenges compared to binary classification. A skewed class distribution often reduces the effectiveness of machine learning algorithms, particularly in predicting samples from minority classes.

In this question, you will perform multiclass logistic regression (softmax classification) on the *20 Newsgroups* dataset (available at `https://www.kaggle.com/datasets/crawford/20-newsgroups`).

1. Since the original dataset is balanced, preprocess the data to create an imbalanced version by removing a subset of news articles from selected classes. An example of such an imbalance is shown in the dataset description.

2. Perform multiclass classification using logistic regression on both the balanced and imbalanced versions of the dataset.

3. Evaluate and compare the performance of the classifier in each case by computing the confusion matrix and classification accuracy.

4. You may refer to the following resource for a better understanding of multiclass logistic regression implemented from scratch:
`https://towardsdatascience.com/multiclass-logistic-regression-from-scratch-9cc0007da372`

Q4. Given a set of face images from the ORL face database, use Principal Component Analysis (PCA) to perform dimensionality reduction. You may use inbuilt functions for computing the principal components. Perform the following steps:

1. Use the ORL database consisting of $N = 400$ face images of size 112×92, resulting in a data matrix $X \in \mathbb{R}^{10304 \times 400}$. Display a few face images to visualize the dataset.

2. Mean-subtract the data and compute the sample covariance matrix

$$S \in \mathbb{R}^{400 \times 400}.$$

3. Perform eigendecomposition of the covariance matrix S to obtain the matrix U, where each column of U corresponds to an eigenvector.

4. Each eigenvector represents an *eigenface*. Reshape each eigenvector to match the original image size and visualize the eigenfaces for

$$k = 1, 100, \ldots, D.$$

5. Plot the variance explained by the principal components as a function of the component index k, where $k = 1$ corresponds to the eigenvector with the highest variance. Consider $k = 1, 100, \ldots, D$.

6. Use the first k eigenvectors (principal components) to obtain the low-dimensional representation of the data:

$$V = U^T X.$$

Vary k from 1 to D in steps of 100.

7. Compute the reconstruction error between the original data X and the reconstructed data $\hat{X}$ using the Frobenius norm (or matrix norm), defined as the square root of the sum of squared reconstruction errors.

8. Plot the reconstruction error as a function of the number of principal components k. Consider $k = 1, 100, \ldots, D$.

9. Visualize the reconstructed face images and compare them with the original images.

10. Determine the most suitable value of k. Report the corresponding reconstruction error and loss in variance. Display 10 original face images along with their corresponding reconstructed images for the selected value of k.